

Analysis of Complex Sample Survey Data

SURV/SURVMETH 701

Lecture Notes: Module 1

Complex Sample Designs and Design Effects in Survey Estimation and Inference

Instructor: Yan Li

e-mail: yli6@umd.edu

Analysis of Complex Sample Survey Data: Course Objectives

1. Develop a comprehensive understanding of design-based estimation and inference for complex sample survey data.
2. Acquire the ability to apply the theory to the range of problems encountered in survey practice.
3. Learn how to use software tools that are available for analysis of complex survey data.
4. Gain an understanding of how to interpret the results of analysis of complex sample data, including robustness and sensitivity.
5. Practice presentation of survey results and discussion of survey findings through homework and exams.

Steps in Applied Survey Data Analysis

Step	Activity
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- | | |
|---|--|
| 1 | Defining the analysis objectives. <ul style="list-style-type: none">• Describe characteristics of a target population• Explore and extend the understanding of multivariate relationships among variables in the target population |
| 2 | Understanding the sample design. <ul style="list-style-type: none">• These design features may require special approaches to population estimation and inference• Chapters 2-4 fundamental features of sample designs• Chapters 5-12 simple extensions of applied stat methods |
| 3 | Understanding design variables, underlying constructs, and missing data. <ul style="list-style-type: none">• The key features of sample design – design variables |

Steps in Applied Survey Data Analysis

Step	Activity
4.	Analyzing the data. • Univariate descriptive and simple bivariate analyses of continuous and categorical variables • Linear regression model for continuous dependent variable • Generalized linear regression modeling methods • Cox proportional hazard model and discrete time logistic models • Methods for missing data problems
5	Interpreting and evaluating the results of the analysis.
6	Reporting of estimates and inferences from the survey data. • Communicate effectively the findings to fellow scientists, policy analysts and administrators and decision makers

Steps in Applied Survey Data Analysis

Step	Activity
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- | | |
|---|---|
| 1 | Defining the analysis objectives. <ul style="list-style-type: none">• ALAD G177C single nucleotide polymorphism (SNP) affects the relationship between lead and mortality |
| 2 | Understanding the sample design. <ul style="list-style-type: none">• Data: National Health and Nutrition Examination Survey (NHANES III)• What are the sample designs of NHANES III? |
| 3 | Understanding design variables, underlying constructs, and missing data. <ul style="list-style-type: none">• Design variables: stratification variable, PSU, sample weights, age, race• Distribution of Risk factors, such as smoking and alcohol consumption?• Missing data pattern? |

Steps in Applied Survey Data Analysis

Step	Activity
4.	Analyzing the data. • Cox proportional hazards regression • Estimation of the relative risk of mortality by ALAD genotype, and by blood lead levels ($<5 \mu\text{g/dL}$ vs. $\geq 5 \mu\text{g/dL}$), and their interaction.
5	Interpreting and evaluating the results of the analysis. • The adjusted overall relative risk for participants with the variant ALAD genotype was decreased for all-cause mortality (hazards ratio = 0.68; [95% confidence interval = 0.50-0.93]) compared with persons having the common GG genotype. There was some suggestion that higher lead levels were associated with cancer mortality (1.48 [0.92-2.38]). We observed no convincing interaction effect between ALAD genotype and blood lead level on mortality risk.
6	Reporting of estimates and inferences from the survey data.

ASDA Web Site:

<http://www.isr.umich.edu/src/smp/asda/>

Applied Survey Data Analysis

[Project Overview](#) [Information about Authors](#) [Links to Data Sets for First and Second Editions](#) [Links to Additional Sites](#)

[Survey Data Analysis Publications](#) [Professional Reviews](#) [Frequently Asked Questions](#) [Supplemental Code Examples](#)

Site Overview

This site contains information about the text "*Applied Survey Data Analysis*", (first and second editions) including author biographies, links to public release data sets and related sites, code and output for analysis examples replicated in current software packages, and information about new publications of interest to survey data analysts. Other features include a FAQ log and links to other software and statistical sites. We plan to intermittently update this site with news about ongoing statistical and software advances in the field of analysis of survey data.

Special Notes from Authors

ASDA-Second Edition is Available as of June 28, 2017!

Project Overview

Applied Survey Data Analysis is the product born of many years of teaching applied survey data analysis classes and practical experience analyzing survey data. We have taught various versions of this course in the ISR/SRC Summer Institute Program, as part of University of Michigan/CSCAR, and within the Survey Methodology Program at University of Michigan and University of Maryland. Our goal has been to integrate teaching materials and practical analysis knowledge into a textbook geared to a level accessible for graduate students and working analysts who may have varying levels of statistical and analytic expertise. We intend to update the materials on this website as statistical and software improvements emerge with the goal of assisting analyst and researchers performing survey data analysis.

Information About Authors

Patricia A. Berglund is a Senior Research Associate in the Survey Methodology Program at the Institute for Social Research. She has extensive experience in the use of computing systems for data management and complex sample survey data analysis. She works on research projects in youth substance abuse, adult mental health, and survey methodology using data from Army STARRS, Monitoring the Future, the National Comorbidity Surveys, World Mental Health Surveys, Collaborative Psychiatric Epidemiology Surveys, and various other national and international surveys. In addition, she is involved in development, implementation, and teaching of analysis courses and computer training programs at the Survey Research Center-Institute for Social Research. She also lectures in the SAS® Institute-Business Knowledge Series. mailto:pberg@umich.edu

Steven G. Heeringa is a Research Scientist in the Survey Methodology Program, the Director of the Statistical and Research Design Group in the Survey Research Center, and the Director of the Summer Institute in Survey Research Techniques at the Institute for Social Research. He has over 25 years of statistical sampling experience directing the development of the SRC National Sample design, as well as sample designs for SRC's major longitudinal and cross-sectional survey programs. During this period he has been actively involved in research and publication on sample design methods and procedures such as weighting, variance estimation, and the imputation of missing data that are required in the analysis of sample survey data. He has been a teacher of survey sampling methods to U.S. and international students and has served as a sample design consultant to a wide variety of international research programs based in countries such as Russia, the Ukraine, Uzbekistan, Kazakhstan, India, Nepal, China, Egypt, Iran, and Chile. mailto:sheering@umich.edu

Brady T. West Brady T. West is a Research Associate Professor in the Survey Methodology Program, located within the Survey Research Center at the Institute for Social Research on the University of Michigan-Ann Arbor (U-M) campus. He also serves as a Statistical Consultant on the U-M Consulting for Statistics, Computing, and Analytics Research (CSCAR) team. He earned his PhD from the Michigan Program in Survey Methodology in 2011. Before that, he received an MA in Applied Statistics from the U-M Statistics Department in 2002, being recognized as an Outstanding First-year Applied Masters student, and a BS in Statistics with Highest Honors and Highest Distinction from the U-M Statistics Department in 2001. His current research interests include the implications of measurement error in auxiliary variables and survey paradata for survey estimation, survey nonresponse, interviewer variance, and multilevel regression models for clustered and longitudinal data. He is the lead author of a book comparing different statistical software packages in terms of their mixed-effects modeling procedures (Linear Mixed Models: A Practical Guide using Statistical Software, Second Edition, Chapman Hall/CRC Press, 2014), and he is a co-author of a second book entitled Applied Survey Data Analysis (with Steven Heeringa and Pat Berglund), which was published by Chapman Hall in April 2010 and has a second edition in press that will be available in mid-2017. Brady lives in Dexter, MI with his wife Laura, his son Carter, his daughter Everleigh, and his American Cocker Spaniel Bailey. mailto:bwest@umich.edu

Professional Reviews of ASDA-Second Edition

Review/Summary from Chapman Hall Website

Features

- Bootstrap methods of variance estimation.
- Estimation and inference for specialized functions such as the Gini coefficient and log-linear models.

SAS v9.4 Code and Results

[Overview of SAS Commands](#)

[Chapter 5 Analysis Examples](#)

[Chapter 6 Analysis Examples](#)

[Chapter 7 Analysis Examples](#)

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Show all

ASDA Web Site: Example Code and Results

Analysis Examples Replication-Second Edition

The analysis examples replication materials cover Chapters 5-13 of ASDA Second Edition but not every software package contains all chapters. Lack of a link for a given chapter indicates that this software package does not include the ability to perform this type of analysis technique.

SAS v9.4 Code and Results

- [Overview of SAS Commands](#)
- [Chapter 5 Analysis Examples](#)
- [Chapter 6 Analysis Examples](#)
- [Chapter 7 Analysis Examples](#)
- [Chapter 8 Analysis Examples](#)
- [Chapter 9 Analysis Examples](#)
- [Chapter 10 Analysis Examples](#)
- [Chapter 11 Analysis Examples](#)
- [Chapter 12 Analysis Examples](#)

Stata v14 Code (Results Are Presented Throughout the Book)

- [Overview of Stata Commands](#)
- [Chapter 5 Analysis Examples](#)
- [Chapter 6 Analysis Examples](#)
- [Chapter 7 Analysis Examples](#)
- [Chapter 8 Analysis Examples](#)
- [Chapter 9 Analysis Examples](#)
- [Chapter 10 Analysis Examples](#)
- [Chapter 11 Analysis Examples](#)
- [Chapter 12 Analysis Examples](#)
- [Chapter 13 Analysis Examples](#)

Professional Reviews of ASDA-Second Edition

Review/Summary from Chapman Hall Website

Features

- Bootstrap methods of variance estimation.
- Estimation and inference for specialized functions such as the Gini coefficient and log-linear models.
- Updated approaches to examining model diagnostics, testing goodness of fit, and estimation and display of marginal effects in linear and generalized linear models.
- State-of-the-art methods for analysis of longitudinal survey data.
- Fractional imputation methods for item missing data.
- Enhanced treatment of methods and software for fitting multilevel models, structural equation models and other latent variable models to complex sample survey data.
- Updated review of software packages for the analysis of complex sample survey data.

Summary

Highly recommended by the Journal of Official Statistics, The American Statistician, and other journals, Applied Survey Data Analysis, Second Edition provides an up-to-date overview of state-of-the-art approaches to the analysis of complex sample survey data. Building on the wealth of material on practical approaches to descriptive analysis and regression modeling from the first edition, this second edition expands the topics covered and presents more step-by-step examples of modern approaches to the analysis of survey data using the newest statistical software.

Designed for readers working in a wide array of disciplines who use survey data in their work, this book continues to provide a useful framework for integrating more in-depth studies of the theory and methods of survey data analysis. An example-driven guide to the applied statistical analysis and interpretation of survey data, the second edition contains many new examples and practical exercises based on recent versions of real-world survey data sets. Although the authors continue to use Stata for most examples in the text, they also continue to offer SAS, SPSS, SUDAAN, R, WesVar, iVWare, and Mplus software code for replicating the examples on the book's updated Web site.

Links to Data Sets for First and Second Editions

National Comorbidity Survey-Replication (Collaborative Psychiatric Epidemiology Surveys)

<http://www.icpsr.umich.edu/cpes> (for online documentation tools and data download)

<http://www.hcp.med.harvard.edu/nca> (for NCS-R specific information)

National Health and Nutrition Examination Survey (National Center for Health Statistics)

<http://www.cdc.gov/nchs/>

Health and Retirement Survey (Institute for Social Research-University of Michigan)

<http://hrsonline.isr.umich.edu>

INTRODUCTION: COMPLEX SAMPLES

Complex Sample Survey Data: Probability Samples

- **Probability sample design:**
 - Each population element has a known, non-zero selection probability
 - Properly weighted, sample estimates are unbiased or nearly unbiased for the corresponding population statistic.
 - Variance of sample statistics can be estimated from the sample data (measurability)
- **Simple random sample (SRS):** A probability sample in which each element has an independent and equal chance of being selected for observation. Closest population sampling analog to independently and identically distributed (iid) data.

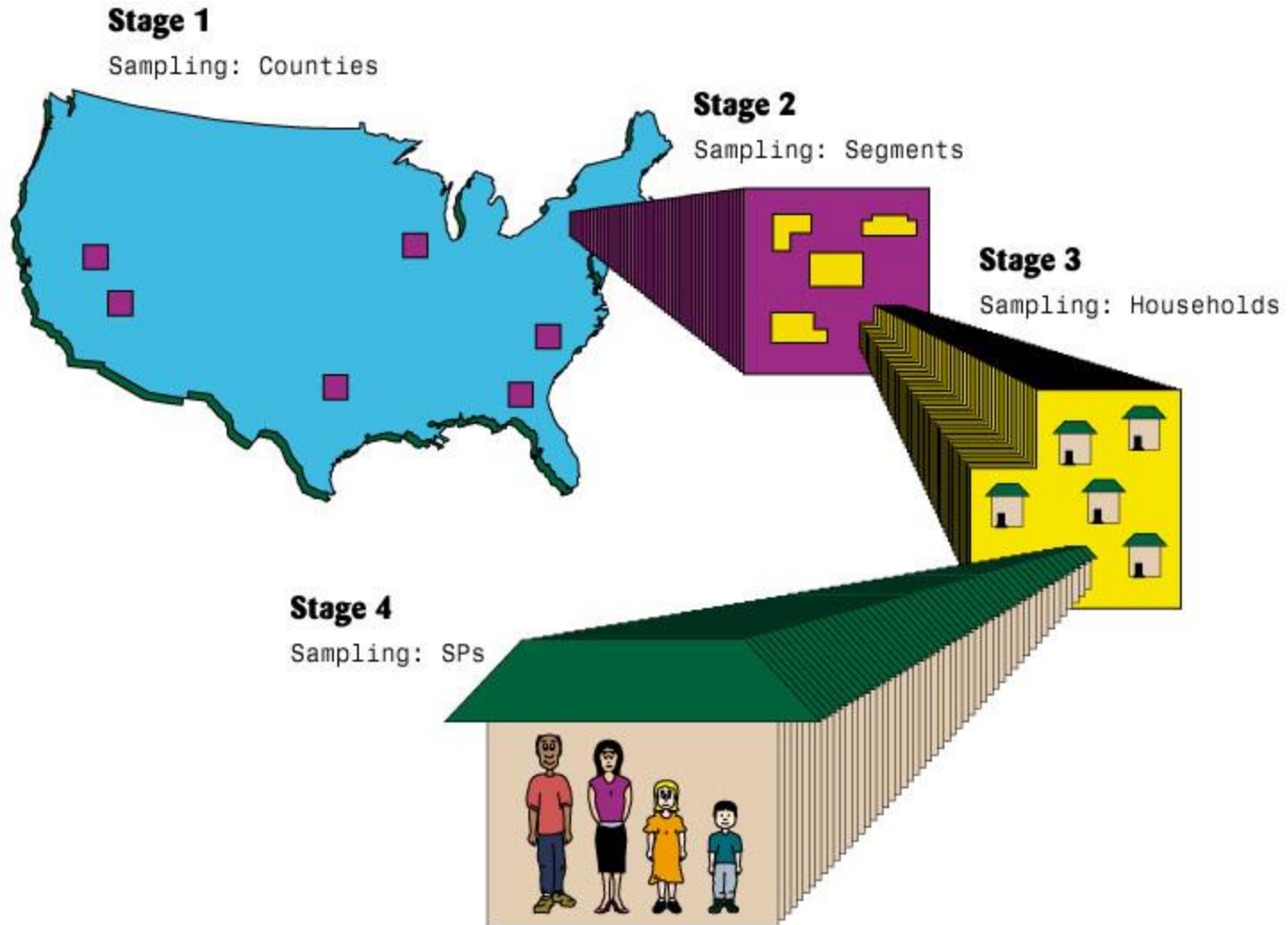
Complex Sample Survey Data: “Complex” Designs

- “Complex sample”:
 - A probability sample developed using sampling procedures such as **stratification, clustering and weighting** designed to improve statistical efficiency, reduce costs or improve precision for subgroup analyses relative to SRS
 - Unbiased estimates with measurable sampling error are still possible
 - Independence and identical distribution of observations (i.i.d.), equal probabilities of selection may no longer hold

Where are Complex Sample Designs Used?

- Complex sample designs are the rule and not the exception in sample-based studies in the Social Sciences, Epidemiology, Public Health, Agriculture, Natural Resources and many other scientific fields
- Common designs
 - Area probability samples of household populations
 - Multi-stage samples of schools, classes and students
 - Stratified samples of businesses, hospitals
- Other designs
 - Dual-frame and other samples in agriculture
 - RDD telephone samples
 - Natural resource samples (time and location samples)

Multi-Stage Sample of U.S. Households: Illustration



NATIONAL HEALTH AND NUTRITION EXAMINATION SURVEY (NHANES)

- NHANES I (1971-75)
- NHANES II (1976-80)
- NHANES III (1988-94)
- NHANES: 1999-present. NHANES is now a continuous survey. New replicate samples every year.
 - ASDA (Second Edition) uses the 2011-2012 data set.

NATIONAL HEALTH AND NUTRITION EXAMINATION SURVEY (NHANES)

- Sampled persons are interviewed about health and health-related matters and undergo medical examinations in mobile examination centers (MECs).
 - <http://www.cdc.gov/nchs/nhanes>
- Separate estimates required for domains defined by the cross-classification of:
 - Age-sex groups, African Americans, Mexican Americans, and All Others.
 - Adolescents, the elderly, Mexican Americans, and the African-American population are oversampled.

NHANES SAMPLE DESIGN

- Primary stage:
 - The PSUs are mostly single counties.
 - Stratify PSUs by region and MSA status.
 - Select PSUs by probability proportionate to estimated size (PPeS) sampling (a few selected with certainty).
- Second Stage:
 - The SSUs are area segments (blocks or combinations of blocks with at least 200-250 persons) within selected PSUs.
 - Area segments are selected with PPeS
- Third Stage:
 - List the housing units in the selected area segments.
 - For each segment, sample housing units from the prepared list.

NHANES: SAMPLING PERSONS WITHIN HOUSEHOLDS

- Conduct a screening interview to classify persons by domain.
- Subsample persons to give required sampling rates by domain for sample persons (SPs). Survey weights will reflect this subsampling step
- Conduct interviews and medical examinations with SPs.
 - A modest percentage of persons will complete the interview but not the medical exam.
 - In NHANES II, $n=25,286$ SPs interviewed, $n^*=20,322$ completed exams
 - Two weights: `fwgthist` (interview data), `fwgtexam` (exam data)

Classification of Sample Designs for Survey Data: Hansen, Madow and Tepping (1983)

Sampling Plan	Method of Inference	
	Design-based	Model-based
Probability Sample	A**	B
Model-dependent Sample	C	D

Sampling Design for Population Studies

- **Sample Plan** – framework and method for selecting the observational units
 - *Probability sampling*: Hansen, Hurwitz, Madow (1953); Kish (1965)
 - *Model-assisted sampling*: Sarndal, Swensson, Wretman (1992)
- **Estimation and Inference**
 - *Design-based* (repeated sampling - **focus of this course**)

We think conceptually about selecting samples over and over according to a specified design. An estimator (for population total, say) is evaluated from each sample. The properties of the estimator are computed by averaging over the set of possible samples Base on the process of selecting the sample
 - *Model-based*

A model is used that is intended to describe the structure that exists in the population. **This structure is the same regardless of how the sample is selected.** Inferences are made using the probability distribution associated with the model

Based on the underlying **structure** in the population

- **Estimation and Inference - *Model-based***
 - *Model-based*
 - Prediction approach (see Valliant, Dorfman, Royall, 2000, *Finite Population Sampling and Inference: A Prediction Approach*, Wiley)
 - Hierarchical Linear Models (see Pfeffermann et al., 1998, “Weighting for unequal selection probabilities in multilevel models,” JRSS-B; Rabe-Hesketh and Skrondal, 2006, “Multilevel Modelling of Complex Survey Data,” JRSS-A)
 - Bayesian Models (Little, 2003, Ch. 4 in *Analysis of Survey Data*)

Statistical Inference (Model-based)

- 1) Incorporates a probability model, $f(y|x, \theta)$ for the variables.
 - a) ex: $Y(1), \dots, Y(n)$ are independent observations from a standard normal distribution with mean μ and variance σ^2 .
 - b) ex: coin flip – Bernouli trial
$$P(\text{Head}) = .50$$
$$T(\text{Tail}) = .50$$
- 2) Goal is to estimate the parameters, θ , of $f(y|x, \theta)$. Assumes no finite limit to the process that generates the sample Y s.
- 3) Estimators, tests of hypothesis, confidence intervals derived on the basis of the probability model.

Extremely powerful if the probability model is correct.

ex:

$$F = \frac{GM_1M_2}{R^2} + \text{error}$$

“Design-Based” Inference from Sample Surveys

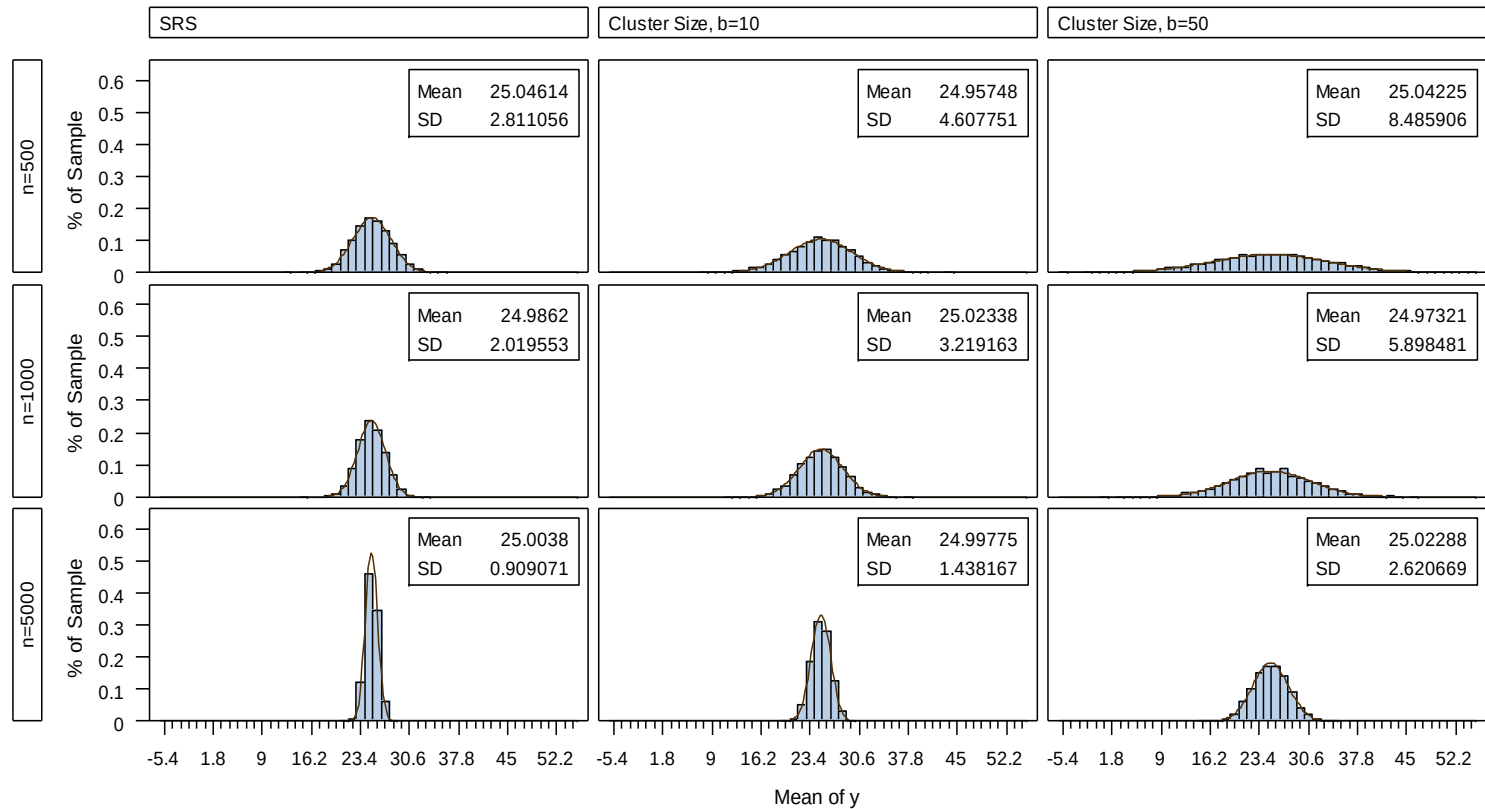
- Goal is inference about finite population parameter θ
- Inference is based on the **expected distribution of sample estimates**, not on a probability model, $f(y|x,\theta)$ for Y s;
- A generic form for confidence interval for a population parameter θ

$$\begin{matrix} (1) & (2) & (3) \\ \hat{\theta} \pm t_{df, 1-\alpha/2} \cdot se(\hat{\theta}) \end{matrix}$$

Example: $\bar{y} \pm t_{\infty, 975} \cdot se(\bar{y})$
 $30.0 \pm 1.96 \times (1.30)$
 $(27.45, 32.54)$

- $t_{df}(p)$ is the value of the Student – t distribution with df degrees of freedom cutting off tail-probability $p/2$. Assumes normality of the sampling distribution of $\hat{\theta}$
- $se(\hat{\theta})$ the estimated standard error (square root of the variance) of the sample estimate, $\hat{\theta}$. Special computational formulas and/or methods are needed.

Sampling Distributions: SRS and Cluster Sampling



Histogram of sample mean with various sample designs

- Sample size
 - $n=500$
 - $n=1000$
 - $n=5000$
- Three levels of clustering
 - No clustering
 - Cluster size $B=10$
 - Cluster size $B=50$
- Observations:
 - Centered at the population mean
 - As n decreases or cluster size B increases, dispersion of sample estimates about the population mean value increases

In real world survey samples, a single sample is observed. If sample size is sufficiently large (e.g. 100 cases) and if the estimate is design-consistent, the sampling distribution converges to an approximately Normal distribution

Concept of the Design Effect

$$D^2(\hat{\theta}) = \frac{SE(\hat{\theta})_{complex}^2}{SE(\hat{\theta})_{srs}^2} = \frac{Var(\hat{\theta})_{complex}}{Var(\hat{\theta})_{srs}}$$

Where:

$D^2(\hat{\theta})$ = the design effect for the sample estimate, $\hat{\theta}$

$Var(\hat{\theta})_{complex}$ = the complex sample design variance of $\hat{\theta}$;

and

$Var(\hat{\theta})_{srs}$ = the simple random sample variance of $\hat{\theta}$

Design Effects for Complex Samples

$$D^2(\hat{\theta}) \approx 1 + f(G_{strat}, L_{cluster}, L_{weighting}),$$

where:

G_{strat} = the relative gain in precision from stratified sampling compared to SRS

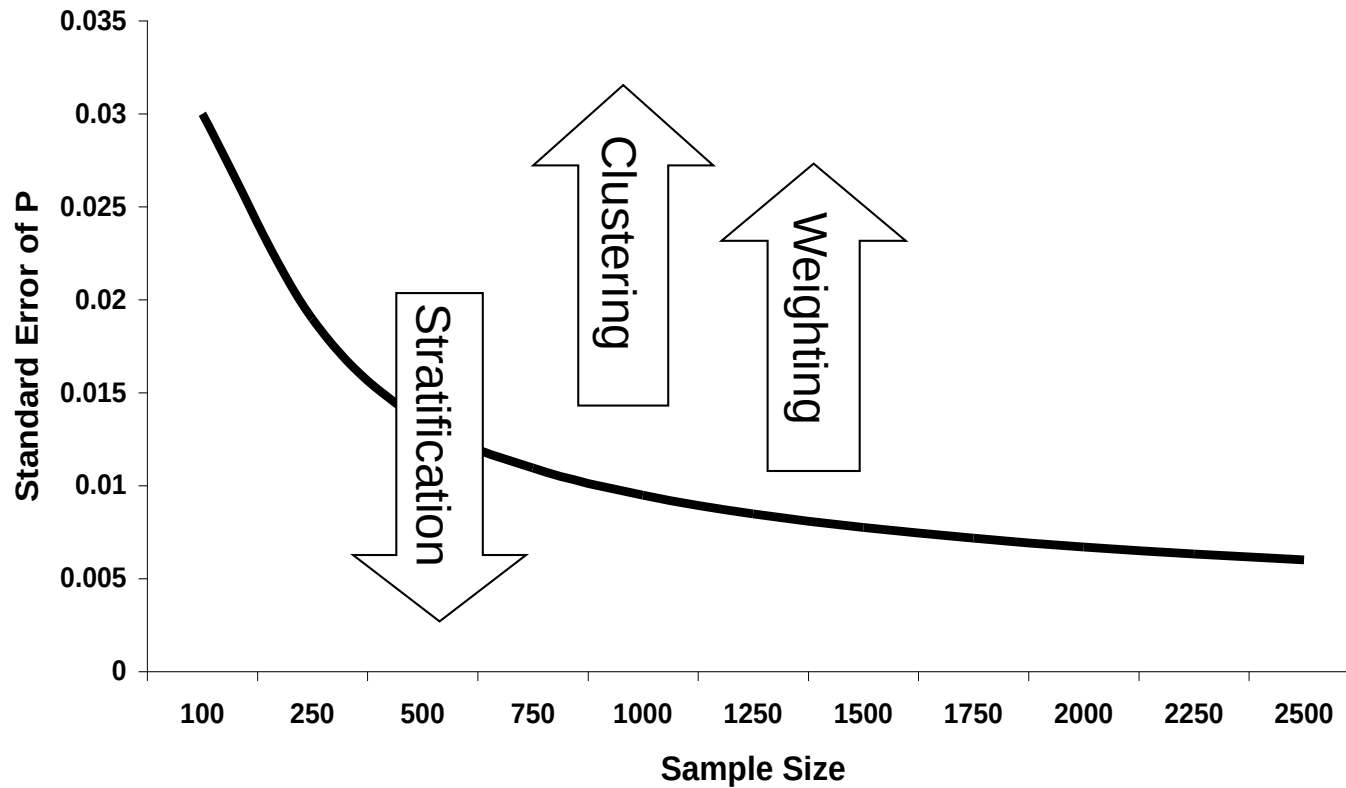
$L_{cluster}$ = the relative loss of precision due to clustered selection of sample elements

$L_{weighting}$ = the relative loss of precision due to unequal weighting for sample elements

“Design” Effects

- ✓ *(Finite population correction)*
- ✓ **Stratification**
- ✓ **Clustering**
- ✓ **Weighting for Selection Probability**
- ✓ *(Nonresponse adjustment, Post-stratification, Raking)*
- ✓ *(Imputation)*

General Influence of Design Features on Variances



Finite Population Correction (*fpc*)

- Sampling without replacement (SWOR)
- *fpc* is the proportion of the population that is not sampled
- *fpc* terms are generally ignored unless explicitly stated
- Simple Random Sampling

$$\text{var}(\bar{y}) = \left(1 - \frac{n}{N}\right) s^2 / n = (1 - f) s^2 / n$$

where: n - sample size
 N - population size

Stratified Random Sampling

$$\text{var}(\bar{y}_{st}) = \sum_{h=1}^H \left(1 - f_h\right) W_h^2 s_h^2 / n_h$$

where: $f_h = n_h / N_h$

Stratified Sampling

Procedure

- form strata
- independent selections in each stratum
- estimate for stratum h: $\bar{y}_h$

- overall estimate: $\bar{y}_{st} = \sum W_h \bar{y}_h$

where $W_h = N_h / N$

- sampling variance: $\sum W_h^2 \text{Var}(\bar{y}_h)$

- estimated sample variance: $\sum W_h^2 \text{var}(\bar{y}_h)$

STRATIFIED SAMPLING - AN EXAMPLE

TWO POPULATION STRATA

Population	Stratum 1 (Urban)	Stratum 2 (Rural)
Size (N) 1,000,000	$N_1 = 200,000$	$N_2 = 800,000$
Variance (S^2) 1,800,000	$S^2_1 = 4,000,000$	$S^2_2 = 1,000,000$
Mean 1,400	3,000	1,000

$< n_1 = 240; n_2 = 960 >$
What is W_h ? and f_h =?

STRATIFIED SAMPLING Example

$$\langle n_1 = 240 \quad n_2 = 960 \rangle$$

$$\begin{aligned} V(\bar{y}_{st}) &= \Sigma(1 - f_h)(W_h^2 S_h^2) / n_h \\ &= (1 - f_1)(W_1^2 S_1^2) / n_1 + (1 - f_2)(W_2^2 S_2^2) / n_2 \\ &\approx (W_1^2 S_1^2) / n_1 + (W_2^2 S_2^2) / n_2 \\ &= (0.2)^2 (4,000,000) / 240 + (0.8)^2 (1,000,000) / 960 \\ &= 666.7 + 666.7 \\ &= 1333 \end{aligned}$$

What if an SRS Was Selected?

$$\langle n = 1,200 \rangle$$

$$\begin{aligned} V(\bar{y}) &= (1 - f)S^2 / n \\ &\approx S^2 / n \\ &= \frac{1,800,000}{1200} \\ &= 1500 \end{aligned}$$

DESIGN EFFECT

$$\begin{aligned} D^2(\bar{y}) &= V(\bar{y}_{st}) / V(\bar{y}_{srs}) \\ &= 1333 / 1500 = 0.89 \end{aligned}$$

$$D^2(\bar{y}) \text{ estimated by } d^2(\bar{y}) = v(\bar{y}_{st}) / v(\bar{y}_{srs})$$

<ALTERNATIVE INTERPRETATION> EFFECTIVE SAMPLE SIZE

What sample size with SRS would be necessary to achieve the same precision (variance) as the given design?

Effective Sample Size = $n / (\text{Design Effect})$

For our example:

$$\text{Effective } n = \frac{1200}{0.89} = 1348$$

Therefore, 1348 sample units need to be selected with SRS in order to achieve the same precision (variance) as the given stratified SRS with stratum defined by urban and rural area.

Gains in Precision Through Stratified Sampling

$$\begin{aligned}\Delta &= \text{var}(\bar{y}_{srs}) - \text{var}(\bar{y}_{st,pr}) \\ &= \frac{S_{total}^2}{n} - \frac{S_{within}^2}{n} \\ &= \frac{S_{within}^2 + S_{between}^2}{n} - \frac{S_{within}^2}{n} \\ &= \frac{S_{between}^2}{n} \\ &= \frac{\sum_{h=1}^H W_h (\bar{Y}_h - \bar{Y})^2}{n}\end{aligned}$$

More variability of the stratum means → More gains in precision

Cluster Sampling

Many populations are widely distributed geographically. We cannot afford visits to n units drawn randomly from the entire area.

Cluster sampling reduces the costs of data collection.

- Sample schools and children within them
- Sample blocks and households within them

Cluster sampling is also useful when the sampling frame lists clusters and not elements. In such a case, select clusters and list elements in selected clusters from which a sample of elements may be drawn:

- Frame of blocks; list households within selected blocks

Cluster Sampling Examples (Complete Clusters)

For the sake of simplicity, we assume that each of the $A = 500$ blocks in the town has exactly $B = 24$ children. We select a SRS of $a = 10$ blocks and examine the immunization history of all children in the selected blocks. The sample consists of $n = aB = 240$ children.

We refer to the A clusters as *primary sampling units*, or PSUs. For each of the $a = 10$ PSUs that are in the sample, we know exactly the proportion of children immunized. Suppose the following are those proportions:

$$y_{\alpha\beta} = \begin{cases} 1 & \text{if immunized} \\ 0 & \text{if not} \end{cases}$$

$$p_{\alpha} = \Sigma y_{\alpha\beta} / B:$$

$$\frac{9}{24}, \frac{11}{24}, \frac{13}{24}, \frac{15}{24}, \frac{16}{24}, \frac{17}{24}, \frac{18}{24}, \frac{20}{24}, \frac{20}{24}, \frac{21}{24}$$

CLUSTER SAMPLE DESIGN EFFECT

- Suppose that a SRS of $n = 240$ children from the town produced the estimate of the proportion immunized as

$$p_0 = 160/240 = 0.667.$$

- Then:
$$v(p_0) = (1 - f) \frac{p_0 q_0}{n - 1} = 0.0009112.$$
- For the cluster sample $v(p_c) = 0.002760$ (quiz).
- The estimated variance of p with the cluster sample is thus larger than the variance of p with a SRS of the same size.

CLUSTER SAMPLE DESIGN EFFECT

- The estimated design effect for the proportion of children immunized is:

$$d^2(p_c) = \frac{v(p_c)}{v(p_0)} = \frac{0.002760}{0.0009112} = 3.029.$$

EFFECTIVE SAMPLE SIZE

- In the example,

$$n_{eff} = \frac{n}{d^2(p_c)}$$

$$n_{eff} = \frac{240}{3.029} = 79.$$

The actual sample size of 240 under clustering design is thus equal to an effective sample size of only 79 with SRS.

DESIGN EFFECT FOR A CLUSTER SAMPLE

- The design effect for the mean of a cluster sample depends on the cluster size (B) and the homogeneity of the elements within the clusters, measured by the intra-class correlation (ρ):

$$D^2(\bar{y}) = 1 + (B - 1)\rho.$$

- For estimates of means or proportions, the intra-class correlation may be estimated by:

$$\hat{\rho} = \frac{d^2(\bar{y}) - 1}{B - 1}.$$

- Thus

$$\hat{\rho} = \frac{3.091 - 1}{24 - 1} = 0.091.$$

INTRA-CLASS CORRELATION ρ

- ρ is a property of the naturally occurring clusters and of the variable under study.
- Since elements in a cluster tend to be similar to one another, ρ is virtually always positive. For human populations, a positive ρ may be due to:
 - *Environment*: exposure to the same climate, hazardous waste, or infectious disease.
 - *Self selection*: wealthy households tend to reside in wealthy neighborhoods and poor households in poor neighborhoods.
 - *Interaction*: shared attitudes among neighbors.
- The magnitude of ρ depends on:
 - The variable under study (e.g., newspaper read, age).
 - The nature of the clusters (e.g., households, city blocks, schools).
 - The size of the clusters (e.g., blocks, enumeration districts, counties).

TWO-STAGE SAMPLING

- Sample a blocks and b children from each block. Blocks are termed primary sampling units (PSUs).
- Consider various combinations of a and b , all with $n = ab = 240$. All these designs are *epsem*, with

$$P(\alpha\beta) = P(\alpha)P(\beta|\alpha) = \frac{a}{500} \times \frac{b}{24} = \frac{240}{500 \times 24} = \frac{1}{50}$$

- The design effect for the two-stage sample design is

$$d^2(p_c) \approx 1 + (b - 1)\hat{\rho}$$

DESIGN EFFECTS FOR TWO-STAGE SAMPLES

a	b	$d^2(p_c) = 1 + (b - 1)\hat{\rho}$	n_{eff}
10	24	$1 + (23 \times 0.091) = 3.03$	79
20	12	$1 + (11 \times 0.091) = 1.97$	122
30	8	$1 + (7 \times 0.091) = 1.62$	148
40	6	$1 + (5 \times 0.091) = 1.44$	167
80	3	$1 + (2 \times 0.091) = 1.18$	204